1. DATASETS & LEAKAGE CONTROL

Public datasets

- **D1** CSE-CIC-IDS2018
- **D2** UNSW-NB15
- **D3** ToN-IoT
- **D4** BoT-IoT (transfer)
- **D5** Lycos2017

Flow timestamps

Used only for ordering and splitting

Output: leakage-controlled windows per dataset

Leakage control

Drop identifiers
IPs, ports, DNS IDs, TTL

Train-only scaling
Median impute, log(1+x), z-score; 41 common features

Time blocks
1-hour (all); 10-min (D5 purged split)

Windows
T = 32, stride 16

SHA-256 deduplication
Remove duplicate windows

Integrity checks
Disjoint splits, class ratios, finite values

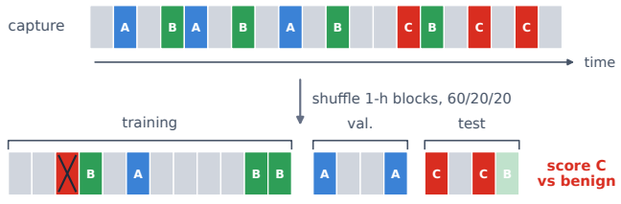
2. PAIRED ZERO-DAY PROTOCOLS

Primary: (1) LOACO vs (2) natural novelty, same data, detectors, seeds

Stress tests: (3) purged blocks · (4) rolling cuts · (5) time split · (6) transfer

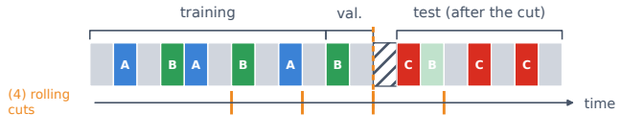
(1) LOACO, grouped_random split

grey = benign; A–C = attack classes



× class C removed from training and validation; test-period benign traffic stays in training

(2) Natural novelty, temporal split



C first seen after the cut (orange) → novel, scored vs benign

B also in training → not scored (faded); val. = last 20% of training period

(3) Purged-block LOACO (short captures, D5)

10-min blocks → **60 / 20 / 20** → Purge adjacent blocks → Min. test windows

3. DETECTOR PANEL & MULTI-SEED EVALUATION

12 detectors / 5 families

- **Confidence** — MSP, Energy, P(attack)
- **Representation** — Stress, Mahalanobis, kNN
- **Benign-only** — IF, OCSVM, PCA, AE
- **Contrastive** — CLAD
- **Reference** — XGBoost, XGBoost-s

Recurrent backbones
(confidence and representation)

- AHSD
- GRU

Multi-seed evaluation

5 seeds → **210** runs → **1** code commit → **Pre-** registered

7 hashed amendments; 4 failed gates reported

4. EVIDENCE & ACCEPTANCE

Metrics

AUC | TPR at 1% FPR | PR-AUC | Seed s.d. | Family ranking | Checkpoint sensitivity

Three main findings

Protocol changes the winner
LOACO winner differs from natural novelty

High AUC, most attacks missed
AUC ≤ 0.75: ≤ 18% caught at 1% FPR

One run is unreliable
Seed variation up to 17×

HoNE acceptance protocol: 8 checks

- ✓ Leakage controls
- ✓ Benign-only + XGBoost
- ✓ Natural novelty + LOACO
- ✓ Operating-point metrics
- ✓ ≥ 5 seeds
- ✓ Checkpoint sensitivity
- ✓ Valid novelty scores
- ✓ Pre-registration / log

Operator review → **TRUST** | **REQUEST EVIDENCE**